

Bisol F.C. Tablets 5 mg

Bisoprolol Fumarate is a synthetic beta1-selective (cardioselective) adrenoceptor blocking agent. The chemical name for Bisoprolol Fumarate is $(\pm)$ -1-[4-[[2-(1-Methylethoxy)ethoxy]methyl]phenoxy]-3-[(1-methylethyl)amino]-2-propanol(E)-2-butenedioate (2:1) (salt). It possesses an asymmetric carbon atom in its structure and is provided as a racemic mixture. The S(-) enantiomer is responsible for most of the beta-blocking activity. Its empirical formula is $(C_{18}H_{31}NO_4)_2 \cdot C_4H_4O_4$.

INGREDIENT

Each F.C. Tablet contains :
Bisoprolol Fumarate..... 5 mg

CLINICAL PHARMACOLOGY

Bisoprolol Fumarate is a beta1-selective (cardioselective) adrenoceptor blocking agent without significant membrane stabilizing activity or intrinsic sympathomimetic activity in its therapeutic dosage range.

Cardioselectivity is not absolute however and at higher doses (20 mg) Bisoprolol Fumarate also inhibits beta2-adrenoceptors, chiefly located in the bronchial and vascular musculature; to retain selectivity it is therefore important to use the lowest effective dose.

Pharmacokinetics and Metabolism

The absolute bioavailability after a 10 mg oral dose of Bisoprolol Fumarate is about 80%. Absorption is not affected by the presence of food. The first pass metabolism of Bisoprolol Fumarate is about 20%.

Binding to serum proteins is approximately 30%. Peak plasma concentrations occur within 2-4 hours of dosing with 5 to 20 mg and mean peak values range from 16 ng/mL at 5 mg to 70 ng/mL at 20 mg. Once daily dosing with Bisoprolol Fumarate results in less than twofold intersubject variation in peak plasma levels. The plasma elimination half-life is 9-12 hours and is slightly longer in elderly patients, in part because of decreased renal function in that population. Steady state is attained within 5 days of once daily dosing. In both young and elderly populations, plasma accumulation is low; the accumulation factor ranges from 1.1 to 1.3 and is what would be expected from the first order kinetics and once daily dosing. Plasma concentrations are proportional to the administered dose in the range of 5 to 20 mg. Pharmacokinetic characteristics of the two enantiomers are similar.

Bisoprolol Fumarate is eliminated equally by renal and non-renal pathways with about 50% of the dose appearing unchanged in the urine and the remainder appearing in the form of inactive metabolites. In humans, the known metabolites are labile or have no known pharmacologic activity. Less than 2% of the dose is excreted in the feces. Bisoprolol Fumarate is not metabolized by cytochrome P450 II D6 (debrisoquin hydroxylase).

In subjects with creatinine clearance less than 40 mL/min, the plasma half-life is increased approximately threefold compared to healthy subjects.

In patients with cirrhosis of the liver, the elimination of Bisoprolol Fumarate is more variable in rate and significantly slower than that in healthy subjects, with plasma half-life ranging from 8.3 to 21.7 hours.

Pharmacodynamics

The most prominent effect of Bisoprolol Fumarate is the negative chronotropic effect resulting in a reduction in resting and exercise heart rate. There is a fall in resting and exercise cardiac output with little observed change in stroke volume, and only a small increase in right atrial pressure, or pulmonary capillary wedge pressure at rest or during exercise.

Findings in short-term clinical hemodynamics studies with Bisoprolol Fumarate are similar to those observed with other beta-blocking agents.

The mechanism of action of its antihypertensive effects has not been completely established. Factors which may be involved include:

- 1) Decreased cardiac output,
- 2) Inhibition of renin release by the kidneys,
- 3) Diminution of tonic sympathetic outflow from the vasomotor centers in the brain.

In normal volunteers Bisoprolol Fumarate therapy resulted in a reduction of exercise- and isoproterenol-induced tachycardia. The maximal effect occurred within 1-4 hours post-dosing. Effects persisted for 24 hours at doses equal to or greater than 5 mg.

Electrophysiology studies in man have demonstrated that Bisoprolol Fumarate significantly decreases heart rate, increases sinus node recovery time, prolongs AV node refractory periods, and, with rapid atrial stimulation, prolongs AV nodal conduction.

Beta1-selectivity of Bisoprolol Fumarate has been demonstrated in both animal and human studies. No effects at therapeutic doses on beta2-adrenoceptor density have been observed. Pulmonary function studies have been conducted in healthy volunteers asthmatics and patients with chronic obstructive pulmonary disease (COPD). Doses of Bisoprolol Fumarate ranged from 5 to 60 mg, atenolol from 50 to 200 mg, metoprolol from 100 to 200 mg, and propranolol from 40 to 80 mg. In some studies, slight, asymptomatic increases in airways resistance (AWR) and decreases in forced expiratory volume (FEV1) were observed with doses of Bisoprolol Fumarate 20 mg and higher, similar to the small increases in AWR also noted with the other cardioselective beta-blockers. The changes induced by beta-blockade with all agents were reversed by bronchodilator therapy.

Bisoprolol Fumarate had minimal effect on serum lipids during antihypertensive studies. In U.S. placebo-controlled trials, changes in total cholesterol averaged +0.8% for Bisoprolol Fumarate-treated patients, and +0.7% for placebo. Changes in triglycerides averaged +19% for Bisoprolol Fumarate-treated patients, and +17% for placebo.

Bisoprolol Fumarate has also been given concomitantly with thiazide diuretics. Even very low doses of hydrochlorothiazide (6.25 mg) were found to be additive with Bisoprolol Fumarate in lowering blood pressure in patients with mild-to-moderate hypertension.

INDICATIONS AND USAGE:

High blood pressure (hypertension).

Coronary heart disease (angina pectoris).

Treatment of stable chronic moderate to severe heart failure with reduced systolic ventricular function (ejection fraction maximum 35%, based on echocardiography) in addition to ACE inhibitors, and diuretics and optionally cardiac glycosides.

CONTRAINDICATIONS:

Bisoprolol Fumarate is contraindicated in patients with cardiogenic shock, overt cardiac failure, second or third degree AV block, and marked sinus bradycardia.

WARNINGS:

Cardiac Failure

Sympathetic stimulation is a vital component supporting circulatory function in the setting of congestive heart failure, and beta-blockade may result in further depression of myocardial contractility and precipitate more severe failure. In general, beta-blocking agents should be avoided in patients with overt congestive failure. However, in some patients with compensated cardiac failure it may be necessary to utilize them. In such a situation, they must be used cautiously.

In Patients Without a History of Cardiac Failure

Continued depression of the myocardium with beta-blockers can, in some patients, precipitate cardiac failure. At the first signs or symptoms of heart failure, discontinuation of Bisoprolol Fumarate should be considered. In some cases, beta-blocker therapy can be continued while heart failure is treated with other drugs.

Abrupt Cessation of Therapy

Exacerbation of angina pectoris and, in some instances, myocardial infarction or ventricular arrhythmia, have been observed in patients with coronary artery disease following abrupt cessation of therapy with beta-blockers. Such patients should, therefore, be cautioned against interruption or discontinuation of therapy without the physician's advice. Even patients without overt coronary artery disease, it may be advisable to taper therapy with BISOPROLOL FUMARATE over approximately one week with the patient under careful observation. If withdrawal symptoms occur, BISOPROLOL FUMARATE therapy should be reinstituted, at least temporarily.

Peripheral Vascular Disease

Beta-blockers can precipitate or aggravate symptoms of arterial insufficiency in patients with peripheral vascular disease. Caution should be exercised in such individuals.

Branchospastic Disease

PATIENTS WITH BRONCHOSPASTIC DISEASE SHOULD, IN GENERAL, NOT RECEIVE BETA-BLOCKERS. Because of its relative beta1-selectivity, however, Bisoprolol Fumarate may be used with caution in patients with bronchospastic disease who do not respond to, or who cannot tolerate other antihypertensive treatment. Since beta1-selectivity is not absolute, the lowest possible dose of Bisoprolol Fumarate should be used, with therapy starting at 2.5 mg. A beta2 agonist (bronchodilator) should be made available.

Anesthesia and Major Surgery

If Bisoprolol Fumarate treatment is to be continued perioperatively, particular care should be taken when anesthetic agents which depress myocardial function, such as ether, cyclopropane, and trichloroethylene, are used. See OVERDOSAGE for information on treatment of bradycardia and hypotension.

Diabetes and Hypoglycemia

Beta-blockers may mask some of the manifestations of hypoglycemia, particularly tachycardia. Nonselective beta-blockers may potentiate insulin-induced hypoglycemia and delay recovery of serum glucose levels. Because of its beta1-selectivity, this is less likely with Bisoprolol Fumarate. However, patients subject to spontaneous hypoglycemia, or diabetic patients receiving insulin or oral hypoglycemic agents, should be cautioned about these possibilities and Bisoprolol Fumarate should be used with caution.

Thyrotoxicosis

Beta-adrenergic blockade may mask clinical signs of hyperthyroidism, such as tachycardia. Abrupt withdrawal of beta-blockade may be followed by an exacerbation of the symptoms of hyperthyroidism or may precipitate thyroid storm

PRECAUTIONS:

Impaired Renal or Hepatic Function

Use caution in adjusting the dose of Bisoprolol Fumarate in patients with renal or hepatic impairment .

Use in Elderly Patients

Bisoprolol Fumarate has been used in elderly patients with hypertension. Response rates and mean decreases in systolic and diastolic blood pressure were similar to the decreases in younger patients in the U.S. clinical studies. Although no dose response study was conducted in elderly patients, there was a tendency for older patients to be maintained on higher doses of Bisoprolol Fumarate.

Observed reductions in heart rate were slightly greater in the elderly than in the young and tended to increase with increasing dose. In general, no disparity in adverse experience reports or dropouts for safety reasons was observed between older and younger patients. Dose adjustment based on age is not necessary.

Pediatric Use

Safety and effectiveness in pediatric patients have not been established.

DRUG INTERACTIONS

Bisoprolol Fumarate should not be combined with other beta-blocking agents. Patients receiving catecholamine-depleting drugs, such as reserpine or guanethidine, should be closely monitored, because the added beta-adrenergic blocking action of Bisoprolol Fumarate may produce excessive reduction of sympathetic activity. In patients receiving concurrent therapy with clonidine, if therapy is to be discontinued, it is suggested that Bisoprolol Fumarate be discontinued for several days before the withdrawal of clonidine. Bisoprolol Fumarate should be used with care when myocardial depressants or inhibitors of AV conduction, such as certain calcium antagonists (particularly of the phenylalkylamine [verapamil] and benzothiazepine [diltiazem] classes), or antiarrhythmic agents, such as disopyramide, are used concurrently. Concurrent use of rifampin increases the metabolic clearance of Bisoprolol Fumarate, resulting in a shortened elimination half-life of Bisoprolol Fumarate. However, initial dose modification is generally not necessary. Pharmacokinetic studies document no clinically relevant interactions with other agents given

concomitantly, including thiazide diuretics, digoxin, and cimetidine. There was no effect of Bisoprolol Fumarate on prothrombin time in patients on stable doses of warfarin.

Risk of Anaphylactic Reaction: While taking beta-blockers, patients with a history of severe anaphylactic reaction to a variety of allergens may be more reactive to repeated challenge, either accidental, diagnostic, or therapeutic. Such patients may be unresponsive to the usual doses of epinephrine used to treat allergic reactions.

PREGNANCY AND LACTATIONS

Pregnancy Category C

In rats, Bisoprolol Fumarate was not teratogenic at doses up to 150 mg/kg/day which is 375 and 77 times the MRHD on the basis of body-weight and body-surface-area, respectively. Bisoprolol Fumarate was fetotoxic (increased late resorptions) at 50 mg/kg/day and maternotoxic (decreased food intake and body weight gain) at 150 mg/kg/day. The fetotoxicity in rats occurred at 125 times the MRHD on a body-weight-basis and 26 times the MRHD on the basis of body-surface-area. The maternotoxicity occurred at 375 times the MRHD on a body-weight basis and 77 times the MRHD on the basis of body-surface-area. In rabbits, Bisoprolol Fumarate was not teratogenic at doses up to 125 mg/kg/day which is 31 and 12 times the MRHD based on body-weight and body-surface-area, respectively, but was embryolethal (increased early resorptions) at 12.5 mg/kg/day. There are no adequate and well-controlled studies in pregnant women. BISOPROLOL FUMARATE should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

Nursing Mothers

Small amounts of Bisoprolol Fumarate (<2% of the dose) have been detected in the milk of lactating rats. It is not known whether this drug is excreted in human milk. Because many drugs are excreted in human milk caution should be exercised when Bisoprolol Fumarate is administered to nursing women.

ADVERSE REACTIONS:

The following is a comprehensive list of adverse experiences reported with Bisoprolol Fumarate in worldwide studies, or in postmarketing experience :

Central Nervous System: Dizziness, unsteadiness, vertigo, syncope, headache, paresthesia, hypoaesthesia, hyperaesthesia, somnolence, sleep disturbances, anxiety/restlessness, decreased concentration/memory.

Autonomic Nervous System: Dry mouth

Cardiovascular: Bradycardia, palpitations and other rhythm disturbances, cold extremities, claudication, hypotension, orthostatic hypotension, chest pain, congestive heart failure, dyspnea on exertion.

Psychiatric: Vivid dreams, insomnia, depression.

Gastrointestinal: Gastric/epigastric/abdominal pain, gastritis, dyspepsia, nausea, vomiting, diarrhea, constipation, peptic ulcer.

Musculoskeletal: Muscle/joint pain, arthralgia, back/neck pain, muscle cramps, twitching/tremor.

Skin: Rash, acne, eczema, psoriasis, skin irritation, pruritus, flushing, sweating, alopecia, dermatitis, angioedema, exfoliative dermatitis, cutaneous vasculitis.

Special Senses: Visual disturbances, ocular pain/pressure, abnormal lacrimation, tinnitus, decreased hearing, earache, taste abnormalities.

Metabolic: Gout

Respiratory: Asthma/bronchospasm, bronchitis, coughing, dyspnea, pharyngitis, rhinitis, sinusitis, URI.

Genitourinary: Decreased libido/impotence, Peyronie's disease, cystitis, renal colic, polyuria.

Hematologic: Purpura.

General: Fatigue, asthenia, chest pain, malaise, edema, weight gain, angioedema.

In addition, a variety of adverse effects have been reported with other beta-adrenergic blocking agents and should be considered potential adverse effects of Bisoprolol Fumarate:

Central Nervous System: Reversible mental depression progressing to catatonia, hallucinations, an acute reversible syndrome characterized by disorientation to time and place, emotional lability, slightly clouded sensorium.

Allergic: Fever, combined with aching and sore throat. laryngospasm, respiratory distress.

Hematologic: Agranulocytosis, thrombocytopenia, thrombocytopenic purpura.

Gastrointestinal: Mesenteric arterial thrombosis, ischemic colitis.

Miscellaneous: The oculomucocutaneous syndrome associated with the beta-blocker practolol has not been reported with Bisoprolol Fumarate during investigational use or extensive foreign marketing experience.

Laboratory Abnormalities

In clinical trials, the most frequently reported laboratory change was an increase in serum triglycerides, but this was not a consistent finding. Sporadic liver test abnormalities have been reported. In the U.S. controlled trials experience with Bisoprolol Fumarate treatment for 4-12 weeks, the incidence of concomitant elevations in SGOT and SGPT from 1 to 2 times normal was 3.9%, compared to 2.5% for placebo. No patient had concomitant elevations greater than twice normal.

In the long-term, uncontrolled experience with Bisoprolol Fumarate treatment for 6-18 months, the incidence of one or more concomitant elevations in SGOT and SGPT of between 1-2 times normal was 6.2%. The incidence of multiple, occurrences was 1.9%. For concomitant elevations in SGOT and SGPT of greater than twice normal, the incidence was 1.5%. The incidence of multiple occurrences was 0.3%. In many cases these elevations were attributed to underlying disorders, or resolved during continued treatment with Bisoprolol Fumarate.

Other laboratory changes included small increases in uric acid, creatinine, BUN, serum potassium, glucose, and phosphorus and decreases in WBC and platelets. These were generally not of clinical importance and rarely resulted in discontinuation of Bisoprolol Fumarate.

As with other beta-blockers, ANA conversions have also been reported on Bisoprolol Fumarate. About 1.5% of patients in long-term studies converted to a positive titer, although about one-third of these patients subsequently reconverted to a negative titer while on continued therapy.

OVERDOSAGE:

The most common signs expected with overdosage of a beta-blocker are bradycardia, hypotension, congestive heart failure, bronchospasm, and hypoglycemia. To date, a few cases of overdose (maximum: 2000 mg) with Bisoprolol Fumarate have been reported. Bradycardia and/or hypotension were noted. Sympathomimetic agents were given in some cases, and all patients recovered.

In general, if overdose occurs, Bisoprolol Fumarate therapy should be stopped and supportive and symptomatic treatment should be provided. Limited data suggest that Bisoprolol Fumarate is not dialyzable. Based on the expected pharmacologic actions and recommendations for other beta-blockers, the following general measures should be considered when clinically warranted:

Bradycardia

Administer IV atropine. If the response is inadequate, isoproterenol or another agent with positive chronotropic properties may be given cautiously. Under some circumstances, transvenous pacemaker insertion may be necessary.

Hypotension

IV fluids and vasopressors should be administered. Intravenous glucagon may be useful.

Heart Block (second or third degree)

Patients should be carefully monitored and treated with isoproterenol infusion or transvenous cardiac pacemaker insertion, as appropriate.

Congestive Heart Failure

Initiate conventional therapy (i.e., digitalis, diuretics, inotropic agents, vasodilating agents).

Bronchospasm

Administer bronchodilator therapy such as isoproterenol and/or aminophylline.

Hypoglycemia

Administer IV glucose.

DOSAGE AND ADMINISTRATION:

Oral administration.

The dose of Bisoprolol Fumarate must be individualized to the needs of the patient. The usual starting dose is 5 mg once daily. In some patients, 2.5 mg may be an appropriate starting dose. If the antihypertensive effect of 5 mg is inadequate, the dose may be increased to 10 mg and then, if necessary, to 20 mg once daily.

Patients with Renal or Hepatic Impairment

In patients with hepatic impairment (hepatitis or cirrhosis) or renal dysfunction (creatinine clearance less than 40 mL/min), the initial daily dose should be 2.5 mg and caution should be used in dosetitation. Since limited data suggest that Bisoprolol Fumarate is not dialyzable, drug replacement is not necessary in patients undergoing dialysis.

Elderly Patients

It is not necessary to adjust the dose in the elderly, unless there is also significant renal or hepatic dysfunction.

Children

There is no pediatric experience with Bisoprolol Fumarate.

SHELF LIFE:

3 Years

STORAGE CONDITION:

Store below 30°C and away from excess heat and moisture.

PACKAGE:

10x10 tablets in PVC/foil blister strip / box
10 x 3 x 10 tablets in PVC/foil blister strip / box

PRODUCT DESCRIPTION:

It is a yellow, biconvex, heart-shaped, film-coated tablet, with a $\frac{ST}{411}$ on one side, and a score on the other side.

REGISTRATION NO.: Taiwan: PM-047538

Manufactured by:

TAIWAN BIOTECH CO., LTD.

22, Chieh-Shou Road, Taoyuan, Taiwan, R.O.C.

Product Registration Holder:

SUNWARD PHARMACEUTICAL SDN BHD

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